

BREAK INTO AEROSPACE

Aviation Safety Engineering

Subdomain Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

Aviation safety engineering is the discipline that stands between the engineering process and catastrophe. It is not a single method or a single regulatory document — it is a systematic framework for ensuring that aircraft systems, organisations, and operations are designed, certified, and managed in ways that keep the probability of catastrophic failure below levels that society has agreed to accept. It is also, unusually among engineering disciplines, a field where the practitioner must be simultaneously rigorous about method and humble about the limits of that method — because the history of aviation accidents is substantially a history of failures that systematic safety analysis predicted were acceptable.

The field spans four distinct but overlapping employer contexts. OEM safety groups — Airbus System Safety, Boeing Safety Engineering, Embraer, Bombardier — employ safety engineers to develop safety cases for new aircraft types and variants, conduct system safety assessments under ARP4761/4754A, and support Type Certification with the airworthiness authority. Regulatory authorities — EASA Flight Standards and Certification, FAA Aircraft Certification Service, CAA, Transport Canada, ANAC — employ safety engineers to evaluate OEM safety cases, develop certification policy, and oversee compliance with airworthiness standards. Airline safety departments employ safety engineers in Safety Management System roles, investigating occurrences, analysing safety data, managing safety risk, and maintaining airworthiness through continued airworthiness programmes. A fourth domain — MRO organisations, avionics manufacturers, and aerostructures suppliers — employs safety engineers in Design Organisation Approval (DOA) and Production Organisation Approval (POA) contexts, ensuring that changes to type-certificated products comply with the safety requirements that governed the original certification.

The technical baseline is shared across all four contexts: the physical and probabilistic reasoning of system safety analysis, the ARP4761/4754A methodology for civil aircraft, the regulatory framework of CS-25 / FAR Part 25 and the supporting AMC/AC material, and the SMS framework that governs safety management in approved organisations. The cultural register, the depth of technical probing, and the career trajectory differ substantially between an OEM system safety engineer who spends their career on a single aircraft programme and an EASA certification engineer who assesses safety cases across dozens of aircraft types simultaneously.

PART I • WHAT AVIATION SAFETY ENGINEERING ACTUALLY LOOKS FOR

By Level

- **Intern / Graduate** — At entry level, aviation safety employers are not primarily screening for knowledge of specific safety methods — FTA and FMEA can be taught, and most graduates who have encountered them in coursework have encountered them superficially. What safety engineering interviews screen for at graduate level is the ability to think systematically about failure — to look at a system and ask “what could go wrong, under what conditions, with what consequence, with what probability?” — and to do so with the discipline and rigour that the field requires rather than the informality that characterises most everyday engineering reasoning.

The distinguishing question at graduate level is not “what is an FTA?” but “I have a dual-channel hydraulic system where each channel has an independent pump, accumulator, and actuator set. Walk me

through how you would construct a fault tree for the loss of all hydraulic actuation to the primary flight controls.” A candidate who can structure this — who identifies the top event, correctly identifies the AND gate structure for the dual-channel redundancy, populates the basic events with failure modes and approximate failure rates, and explains why the independence assumption between the two channels matters for the AND gate probability calculation — is demonstrating systematic safety thinking that immediately distinguishes them from candidates who can only describe the FTA process conceptually.

Graduate interviews at OEM safety groups typically include a structured technical problem — either a simplified fault tree or an FMEA exercise — followed by questions on the regulatory basis for the safety requirements and on behavioural judgement in situations where safety and schedule are in tension. EASA and FAA graduate-entry roles place greater emphasis on regulatory knowledge and the policy reasoning behind airworthiness standards. Airline safety department entry-level roles focus more on SMS concepts, occurrence reporting, and safety data analysis.

- **Mid-Level (Engineer, Senior Engineer, ~3–8 years)** — At mid-level, safety engineering interviews become method-specific and programme-specific. The interviewer is not assessing whether you understand what a fault tree is — they are assessing whether you can construct one correctly for a system of real complexity, handle the common traps (dependent failures, common cause failures, maintenance error combinations), and engage meaningfully with the regulatory standard against which the safety case will be assessed.

Technical interviews at OEM groups probe at the level of specific ARP4761 assessment types: what a Functional Hazard Assessment (FHA) produces and how it feeds a System Safety Assessment (SSA), what the distinction is between a Preliminary System Safety Assessment (PSSA) and a final SSA, what Particular Risks analysis covers, and how Common Cause Analysis — Zonal Safety Analysis (ZSA), Particular Risk Analysis (PRA), and Common Mode Analysis (CMA) — integrates with the quantitative assessments. EASA and FAA interviews at mid-level probe the same methods from the regulator’s perspective: what evidence does a safety case need to contain to be approvable, what are the most common deficiencies in OEM safety submissions, and how does a Means of Compliance differ from a Compliance Verification Document.

The specific context driving mid-level hiring in 2025–2026: the EASA rulemaking activity associated with CS-25 amendments (particularly the ongoing CS-25 SAIL rulemaking for large aeroplanes), the entry into service of new aircraft types (A321XLR, 777X, A220 expansion), the growing demand for safety engineering expertise in electric aircraft and eVTOL certification under SC-VTOL-01 and the FAA’s MOSAIC rulemaking, and the continued expansion of airline Safety Management Systems under ICAO Annex 19 and EU-OPS ORO.GEN.200 are all simultaneously creating demand for mid-level safety engineers with genuine system safety assessment competence.

- **Senior (Principal Safety Engineer, Head of System Safety, Responsible Engineering Authority)** — At senior level, aviation safety engineering interviews are fundamentally about technical authority and the ability to make and defend safety arguments under regulatory scrutiny. The senior safety engineer at an OEM is the person who signs the System Safety Assessment and represents the organisation’s safety position to the airworthiness authority. The senior safety engineer at a regulator is the person who decides whether the OEM’s safety argument is sufficient for type certification. In both cases the role requires not just technical competence but the ability to hold a technically defensible position under challenge from both sides — from the OEM arguing that their design is safe enough, and from within the authority arguing that certification should proceed despite unresolved uncertainties.

The senior interview typically involves a detailed discussion of the most complex safety case the candidate has personally owned, the most significant safety argument they have had to construct and defend, and — specifically at regulatory authorities — the cases where they have had to refuse an OEM’s safety argument and explain why the evidence was insufficient. The story of a refused safety argument is specifically valued at authority interviews because it reveals the candidate’s ability to hold a technically defensible position under commercial pressure.

Hands-On Experience and Domain Knowledge Treated as Strong Signals

Domain	Why It Signals Depth
ARP4761 System Safety Assessment	The primary methodology standard for civil aircraft. FHA identifies failure conditions, PSSA derives safety requirements, SSA verifies implementation. Understanding how Common Cause Analysis integrates with these processes demonstrates specific OEM-tested depth.
ARP4754A Systems Development Process	The companion standard defining development assurance. Understanding how Development Assurance Levels (DAL) flow from the FHA through the PSSA, determining design and verification rigour, distinguishes engineers ready for Type Certification work.
CS-25 / FAR Part 25 Airworthiness	CS-25.1309 requires Catastrophic failure condition probabilities $< 1 \times 10^{-9}/\text{FH}$, Hazardous $< 1 \times 10^{-7}/\text{FH}$, and Major $< 1 \times 10^{-5}/\text{FH}$. Knowing these targets and their rationale is baseline regulatory literacy.
Fault Tree Analysis (FTA)	The primary quantitative method. Correctly identifying minimal cut sets, assigning failure rates from valid sources, bounding the system correctly, and spotting independence traps separates mid-level engineers from beginners.
Failure Modes and Effects Analysis	The bottom-up complement (FMEA/FMES) to FTA. Understanding traceability — how a failure mode maps to an FTA basic event and validates independence assumptions — proves integration capability.
Safety Management Systems (SMS)	Annex 19 mandates SMS. Understanding Safety Risk Management (hazard registers, bow-tie models, risk matrices, proactive safety indicators) is the operational core required for airline and MRO safety roles.
Part-21 DOA & the REA Role	Part-21 Subpart J defines Design Organisation Approval. Knowing Major vs. Minor classifications, AHDC vs. STC, and the personal accountability of the Responsible Engineering Authority (REA) is vital for design change certification.
Continued Airworthiness	Safety continues through the operational life. Understanding how occurrences trigger safety reviews, generating Mandatory Continuing Airworthiness Information (MCAI) and Airworthiness Directives (ADs), is highly valued in oversight roles.
Quantitative Risk Assessment	Recognising that 10^{-9} is an argument of probability and architecture, not a directly verifiable fact. Understanding data sources, uncertainty bounds, and qualitative evidence complements quantitative analysis.
Human Factors in Aviation Safety	Human error contributes to most accidents. Applying the Reason Swiss Cheese model, SHELL, HFACS, and knowing how to incorporate human error probabilities into FTAs demonstrates human-systems integration depth.

How Aviation Safety Engineering Interviews Differ from Generic Systems Engineering Roles

- **The regulatory framework is not background context — it is the primary design constraint.** The safety engineer does not first design a system and then check it against regulations; the safety engineer uses the regulatory methodology (ARP4761, CS-25.1309) to derive requirements that constrain the design.

Treating regulations as mere reference documentation reveals a fundamental misunderstanding.

- **Rigour and conservatism are assessed as professional competencies, not just technical ones.** The cost of being wrong is measured in lives. Interviewers assess the willingness to flag unvalidated assumptions, escalate close-to-limit findings, and resist schedule pressure. Stories showing professional courage are highly valued.
- **The independence assumption is the most commonly probed technical trap.** FTA relies on basic events in AND gates being independent. Common cause failures defeat this. Candidates who cannot explain Common Cause Analysis (CMA, ZSA, PRA) and what evidence validates independence are missing the conceptual centrepiece of safety methodology.
- **Safety cases are arguments, not calculations.** A computed probability of 0.8×10^{-9} might not be approvable if independence assumptions are unvalidated. A safety case is a structured argument combining quantitative analysis, qualitative assessment, Common Cause Analysis, and development assurance.
- **Occurrence investigation is a specific and valued skill, not just a process.** Analysing in-service events requires structured methodologies (Reason's model, HFACS, Bow-Tie) to find root causes without as-signing blame. Many interviews include a mini-investigation exercise.
- **The new entrant challenge is real — and interviewers probe how candidates handle it.** eVTOLs, hydrogen, and highly automated systems challenge assumptions built into CS-25. Senior engineers must be comfortable arguing from first principles when established failure rate data or historical precedents are absent.

PART II · 20 SPECIFIC PREPARATION TASKS

Intern / Graduate

1. Build a fault tree for a real dual-redundant aircraft system — from scratch, with correct gate logic and failure rate estimates 5–7 h

Skill: The primary quantitative safety analysis method, constructed correctly at interview level

Objective: Constructing a fault tree correctly tests simultaneously whether the candidate understands system architecture, probabilistic logic, failure mode reasoning, and independence assumptions. A candidate who places an OR gate where an AND gate belongs has revealed that they do not understand what redundancy achieves probabilistically.

Done Signal: You have constructed a complete fault tree on paper for a dual-redundant system. The tree has a clearly defined top event, correct AND/OR gate assignments, basic events with plausible failure rate estimates, and a computed top event probability. You can identify minimal cut sets, explain which cut set dominates, and describe what the independence assumption requires.

2. Understand the CS-25.1309 safety objectives — probability targets, failure condition severity classifications, and their regulatory basis 3–4 h

Skill: The foundational regulatory requirement of all civil aircraft system safety

Objective: CS-25.1309 targets (10^{-9} /FH for Catastrophic, 10^{-7} /FH for Hazardous, 10^{-5} /FH for Major) derive from an overall fatal accident rate target. Understanding these targets, the five severity classes, and what AMC 25.1309 requires is the regulatory literacy baseline. A candidate who states the target in flights instead of flight hours is missing vital nuance.

Done Signal: You can state the probability targets, explain why flight hours is the reference unit, and explain the rationale connecting these targets to the overall fatal accident rate. You can define each of the five severity classes and give one example of a failure condition in each class for a commercial aircraft.

3–4 h

3. Understand the ARP4761 process structure — FHA, PSSA, SSA, and Common Cause Analysis

4–5 h

Skill: The methodology standard that governs civil aircraft system safety assessment

Objective: ARP4761 defines the structure of every safety case. The FHA identifies failure conditions, PSSA derives safety requirements, and SSA verifies implementation. Common Cause Analysis validates independence assumptions. Candidates must know what each produces and how they interlock.

Done Signal: You can describe the inputs, outputs, and purpose of the Aircraft FHA, System FHA, PSSA, and SSA. You can describe the three components of Common Cause Analysis (ZSA, PRA, CMA) and explain what each checks that the FTA and FMEA individually cannot.

4. Prepare behavioural answers that connect safety rigour to professional courage

2–3 h

Skill: STAR-format stories with the held-position-under-pressure emphasis safety engineering interviews specifically probe

Objective: Interviews probe whether candidates demonstrate professional courage: holding a defensible safety position against schedule pressure, or escalating a concern when informal channels fail. The strongest stories show a safety engineering intervention that drove a specific design, procedural, or organisational change.

Done Signal: You have four STAR stories covering: an identified safety issue others missed; holding a safety position under pressure; a cross-disciplinary safety problem; and an occurrence investigation that changed the recommended corrective action.

5. Understand the Safety Management System framework — the four pillars of ICAO Annex 19 and how they operate in practice

3–4 h

Skill: The organisational safety management framework that governs all approved aviation organisations

Objective: SMS is the operational framework for identifying hazards and managing risk systematically. Understanding how the four pillars (Safety Policy, Risk Management, Assurance, Promotion) work together is essential for airline and MRO safety roles.

Done Signal: You can describe the four components of SMS and what each produces. You can explain the difference between a hazard and a risk, describe how a bow-tie diagram works, and explain the difference between reactive and proactive safety indicators.

6. Get hands-on with safety analysis tools — draw an FTA and FMEA for a real system using industry-standard software

5–7 h

Skill: Tool competence as practical safety analysis capability

Objective: Professional safety analysis uses dedicated software (Isograph, Relyence) to manage complex fault trees, compute minimal cut sets, and propagate failure rates. A candidate who has constructed an FTA in a dedicated tool demonstrates practical engagement beyond paper exercises.

Done Signal: You have constructed a fault tree and corresponding FMEA for a real system using a dedicated safety analysis tool. The fault tree has at least 20 basic events, linked failure rate data, and computed minimal cut sets. You can explain the cut set output and perform sensitivity analysis. 5–7 h

Mid-Level (~3–8 years)

All graduate tasks still apply. Behavioural stories must come from professional contexts involving real safety assessments, real certification programmes, or real in-service safety investigations.

7. Build a thorough working knowledge of Common Cause Analysis — ZSA, PRA, and CMA — and why each catches failure modes that FTA misses 4–5 h

Skill: The safety analysis framework for failures that defeat redundancy

Objective: The most dangerous failures are common cause failures that defeat multiple redundant channels simultaneously. ZSA checks physical proximity threats, PRA checks external physical threats (bird strike, uncontained engine failure), and CMA checks shared design, software, or maintenance errors.

Done Signal: For ZSA, PRA, and CMA: you can describe the method, state the category of failure it addresses, give one example it would catch that an FTA would not, and describe the required documentary evidence. You can explain what “independence” means for a Catastrophic failure condition and the evidence used to validate it.

8. Understand Part-21 Design Organisation Approval — DOA procedures, Major versus Minor classification, and the REA role 4–5 h

Skill: The regulatory framework that authorises OEMs and MROs to certify design changes

Objective: Part-21 Subpart J defines the DOA. Design changes must be classified as Major (EASA involvement) or Minor (DOA approved). The Responsible Engineering Authority (REA) holds personal accountability for the safety case. Understanding these procedures is critical for design change certification.

Done Signal: You can explain the DOA privileges, the criteria for Major vs. Minor changes, and what the certification evidence package for a Major change contains. You can explain the REA role, its personal accountability, and how the sign-off relates to EASA’s approval.

9. Prepare a technically specific story about conducting or reviewing a safety assessment — what you assessed, what you found, and what changed 2 h

Skill: Applied safety assessment engineering at professional depth

Objective: Mid-level interviews are anchored in the experience of conducting real assessments, finding unanticipated issues, and driving design changes. Stories where an analysis revealed a safety requirement gap or independence violation are highly valued because they prove the analysis actually steered the design.

Done Signal: You have a story naming the system and assessment type where your analysis identified a finding unanticipated by the design team. You describe the finding, the argument you made, the discussion generated, and the specific design or procedural change implemented, ending with a traceable safety consequence.

10. Understand continued airworthiness — how in-service safety issues are managed from occurrence to airworthiness directive 3–4 h

Skill: The safety lifecycle beyond certification — managing the fleet through its operational life

Objective: Aircraft operate for decades, and failures occur that were not predicted. The continued airworthiness system handles this through occurrence reporting, OEM analysis, EASA Mandatory Continuing Airworthiness Information (MCAI), and Airworthiness Directives (AD).

Done Signal: You can describe the chain of events from an in-service occurrence to an AD. You can explain what a repetitive task AD requires versus a one-time modification AD, and describe what “equivalent safety” means in the context of an AD compliance alternative.

11. Understand human factors in aviation safety — error models, SHELL, HFACS, and how human error integrates into safety assessment 3–4 h

Skill: The most consistently underweighted factor in aviation accident causation

Objective: Formal methods handle human error poorly, yet it causes the majority of accidents. Understanding the Reason Swiss Cheese model, the SHELL model, and HFACS taxonomy is essential for integrating human error probabilities into safety assessments and investigating maintenance-related occurrences.

Done Signal: You can explain the Reason Swiss Cheese model and its implication for safety barrier design. You can describe the SHELL model’s components and apply it to a scenario. You can explain HFACS and how it is used to classify human error in investigations.

12. Build working knowledge of software safety — DO-178C, DAL assignment, and independence requirements 4–5 h

Skill: The safety assurance framework for airborne software — the largest source of safety-critical complexity

Objective: DO-178C governs software development assurance. DAL A software requires rigorous structural coverage (MC/DC) and strict independence between development and verification teams. Understanding how the DAL assignment flows from the FHA is required literacy for avionics safety.

Done Signal: You can explain how a DAL is derived from failure condition severity. You can describe what DO-178C requires at DAL A for verification and coverage. You can explain what independence means in the DO-178C context and describe what modified condition/decision coverage (MC/DC) is.

13. Prepare for a live safety assessment or investigation exercise under interview conditions 4–6 h

Skill: Structured safety thinking under time pressure

Objective: Interviews commonly include practical exercises: sketching a fault tree, classifying failure conditions, or conducting a mini-investigation. Applying a systematic method visibly, stating assumptions, and identifying analytical uncertainties is specifically what interviewers assess.

Done Signal: You have practised timed exercises: constructing a fault tree in under 20 minutes, classifying failure conditions against CS-25.1309 criteria, and sketching the causal chain for an occurrence using a structured framework. You can explain your methodology out loud.

Senior Level *(Principal Safety Engineer, Head of System Safety, REA)*

All mid-level tasks still apply. Safety assessment knowledge must extend to directing and reviewing assessments conducted by others, not just conducting them. Regulatory knowledge must extend to the AMC and GM level and to the policy reasoning behind the requirements.

14. Prepare to walk through the complete safety case for a system you personally owned — from FHA through SSA through certification closure 4–6 h

Skill: End-to-end safety case ownership at the level a senior safety engineer must demonstrate

Objective: Senior engineers are expected to have owned a complete safety case. This involves the full FHA-PSSA-SSA-Common Cause cycle and the final certification closure with the authority. The specificity of the experience — identifying an independence violation and driving an architectural change — is what defines the senior interview.

Done Signal: For one safety case you personally owned, you can describe without notes: the system, failure conditions, safety requirements, architecture verification, the most significant finding, its resolution, and the evidence package that closed it with the authority. You can quantify the probability and explain the qualitative arguments.

15. Prepare a technically grounded position on safety certification of a novel aircraft type — eVTOL, hydrogen propulsion, or highly automated flight 4–5 h

Skill: Forward-looking safety methodology awareness at senior engineering depth

Objective: Novel architectures challenge existing methodologies. eVTOLs have distributed electric propulsion where single motor failures are non-catastrophic. Hydrogen brings unprecedented failure modes. Highly automated flight challenges human factors assumptions. Senior engineers must engage with these methodology gaps.

Done Signal: You have a 3-minute technical discussion covering a specific methodology challenge for a novel aircraft type. You identify the gap in the existing methodology, describe the approaches being explored, and connect your engineering background to contributing to the solution.

16. Prepare to lead a safety review board discussion — a finding, the evidence, the risk, and the recommendation 4–5 h

Skill: Technical authority in the organisational safety decision process

Objective: Senior safety engineers present findings to Safety Review Boards or CRI panels. The presentation must state the finding, present evidence/uncertainties, assess risk, and make a recommendation. The ability to defend the finding against pushback from programmes and regulators defines technical authority.

Done Signal: You can deliver a 5-minute safety review board presentation for a specific finding. You assess the risk against CS-25.1309 and make a specific recommendation. You can defend the recommendation against three likely challenge questions from the programme team, the authority, and the chief engineer.

17. Understand the regulatory policy process — how airworthiness standards are developed, amended, and how safety engineering feeds rulemaking 3–4 h

Skill: Strategic regulatory awareness at the level of a senior safety engineer working with authorities

Objective: Regulations evolve through the EASA NPA/CRD or FAA NPRM process. Senior engineers must know how to participate in rulemaking and how accident findings translate into regulatory requirements. Knowing what a Certification Review Item (CRI) is and how it interacts with compliance is essential.

Done Signal: You can explain the EASA rulemaking process and identify stakeholder input opportunities. You can describe how one specific accident finding translated into a regulatory change. You can explain what a CRI is and when it is used.

3–4 h

18. Build quantitative estimation fluency for safety — probability arithmetic, sensitivity analysis, and the limits of the numbers

4–5 h

Skill: Senior-speed probabilistic reasoning about safety arguments

Objective: Senior engineers sanity-check probability calculations in real time. Estimating AND gate probabilities, system target failure rates, and sensitivity bounds is critical. Understanding why 0.5×10^{-9} does not mean the system is "twice as safe", and why data uncertainty bounds are massive, is crucial.

Done Signal: You have worked through five probability estimation problems out loud: AND gate combinations, architecture requirements to meet targets, and sensitivity limits. You identify the most load-bearing assumption in each estimate and describe what evidence would validate it.

All Levels

19. Prepare a specific answer to “Why aviation safety engineering — and why this specific domain?” that reflects genuine technical engagement

1–2 h

Skill: Technically specific motivation that distinguishes safety-committed candidates from general aerospace engineers who find safety interesting

Objective: A generic answer falls flat. Employers look for engineers with deep technical opinions: why independence assumptions are hard to validate, why MC/DC coverage exists, or why the Swiss Cheese model is superior to linear causal chains. The answer must engage with a specific active challenge.

Done Signal: You have a 90-second answer naming your target domain, identifying a specific methodology challenge in that domain, explaining what makes it technically difficult, and connecting your background to its solution. The answer cannot be generic.

20. Map your technical background to a specific safety engineering employer type, domain, and regulatory context — and understand the personal liability implications of the REA role

1–2 h

Skill: Precise self-positioning in an employer landscape where the role carries personal regulatory accountability

Objective: OEM safety engineers, EASA certification engineers, and airline SMS engineers have entirely different accountability structures and daily workflows. The REA role specifically carries personal regulatory accountability for a design change. Candidates must demonstrate self-awareness regarding these different employment contexts.

Done Signal: You can state in under 60 seconds: the employer type you are targeting, your primary safety domain, your regulatory familiarity, and two specific technical contributions you bring. If targeting an REA role, you can explain the personal accountability it carries.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	Build a fault tree from scratch — correct gate logic, failure rates, cut sets	All	5–7 h
2	CS-25.1309 — probability targets, severity classes, regulatory basis	All	3–4 h
3	ARP4761 process structure — FHA, PSSA, SSA, Common Cause Analysis	All	4–5 h
4	STAR stories — flagged issue, held position, cross-discipline, investigation	All	2–3 h
5	SMS framework — four pillars of ICAO Annex 19, hazard vs risk, bow-tie	All	3–4 h
6	Safety analysis tools — FTA and FMEA in dedicated software, cut sets	All	5–7 h
7	Common Cause Analysis — ZSA, PRA, CMA — what each catches and why	Mid+	4–5 h
8	Part-21 DOA — procedures, Major/Minor classification, REA role	Mid+	4–5 h
9	Safety assessment story — unexpected finding, design change, traceable outcome	Mid+	2 h
10	Continued airworthiness — occurrence to AD, MCAI process, compliance	Mid+	3–4 h
11	Human factors — Swiss Cheese, SHELL, HFACS, human error in FTA	Mid+	3–4 h
12	Software safety — DO-178C, DAL assignment, independence, MC/DC	Mid+	4–5 h
13	Timed safety exercises — fault tree, failure condition classification	Mid+	4–6 h
14	Complete safety case story — FHA through SSA through certification closure	Senior	4–6 h
15	Novel aircraft type certification challenge — eVTOL, hydrogen, or automation	Senior	4–5 h
16	Safety review board presentation — finding, evidence, risk, recommendation	Senior	4–5 h
17	Regulatory policy process — NPA/NPRM, rulemaking, accident-to-regulation	Senior	3–4 h
18	Probability estimation fluency — AND gates, sensitivity, limits of numbers	Senior	4–5 h
19	“Why aviation safety — why this domain?” — technically specific	All	1–2 h
20	Employer type / domain / regulatory context / REA accountability positioning	All	1–2 h

THE STRUCTURAL REALITY

Aviation safety engineering is the only engineering discipline in commercial aviation whose practitioners are required, by the nature of their work, to hold simultaneously the position of the organisation they represent and the position of those who might be harmed if their work is wrong. The system safety engineer at an OEM is employed by the manufacturer, paid by the manufacturer, and working within the manufacturer's schedule and cost constraints — and is simultaneously professionally accountable for a safety case that certifies to a regulator and, through that regulator, to the travelling public, that the aircraft is safe. The tension between these accountabilities is not theoretical. It has appeared in every major aviation accident investigation involving a system safety failure, and it is the reason that the most valued behavioural story in a safety engineering interview is the one where the candidate held a technically defensible safety position against pressure from inside their own organisation.

The field also carries a specific epistemological challenge that is largely absent from other engineering disciplines: the safety case is an argument about rare events from limited data. The probability targets of CS-25.1309 — 10^{-9} per flight hour for Catastrophic failure conditions — represent one event per billion flight hours. The global commercial aviation fleet accumulates approximately 70 million flight hours per year. This means that a Catastrophic failure condition at the target rate would be expected to occur once every fourteen years across the entire global fleet. The failure rate data used to substantiate the safety case comes from component populations that have never experienced the relevant failure — or, if they have, the sample size is so small that the statistical confidence interval spans several orders of magnitude. Every safety case is, at some level, an argument under profound uncertainty, and the practitioner who does not understand this — who treats the computed 0.7×10^{-9} as a fact rather than a model output — is the most dangerous kind of safety engineer: one who is confident where confidence is not warranted.

The engineers who succeed in this field are those who can be rigorous and humble simultaneously — who can construct and defend a technically sound safety argument while remaining genuinely curious about what the argument assumes and what it cannot see. They use the methodology because it is the best available framework, not because it is perfect. They know the history of accidents that happened to systems that passed safety assessment, and they keep that history present in their work as a permanent reminder that the absence of a finding in the analysis is not the same as the absence of a hazard in the system. And they understand that the most important safety engineering contribution is often not the one that closes a finding in the fault tree, but the one that asks, before the fault tree is drawn, whether the system architecture has correctly bounded the problem it is trying to solve.